Semantic Grounding and Safe Execution for PostGIS Natural-Language-to-SQL

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Abstract

Natural language interfaces to PostGIS must handle challenges that mainstream text-to-SQL benchmarks rarely exercise: geometry-aware schema grounding, OGC topological and metric predicates, coordinate reference systems, `::geography` casting, KNN operators, and safe refusal of destructive or unbounded queries. We present **NL2Semantic2SQL**, a bilingual semantic-grounding and safe-execution framework that couples a semantic layer (aliases, geometry types, SRIDs, spatial operator rules) with an intent-conditioned router, schema-aware prompting, a SQL postprocessor enforcing read-only safety and `LIMIT` injection on large-table previews, and an execution-time self-correction loop. We evaluate on a 125-question Chongqing PostGIS benchmark (85 Spatial + 40 Robustness across six safety categories). The 40-question Robustness suite is the headline primary result: Full reaches 0.975 vs. baseline 0.450 (paired McNemar $b = 0$, $c = 21$, $p < 10^{-4}$), with bounded-answer compliance on OOM Prevention rising from $1/8$ to $7/8$. On Spatial we pool three independent Full runs, mean $\pm$ SD EX$= 0.663 \pm 0.048$ vs. baseline 0.529; majority-vote $p = 0.052$ (marginal). A secondary BIRD warehouse track shows design-set $+0.093$ ($p = 0.0213$, in-sample tuning) but held-out $+0.033$ (n.s., $p = 0.3833$); we report this conservatively. A bilingual human re-audit of a 50-question Chinese BIRD probe yields paired $p = 1.00$, ruling out translation artefacts as the cause of the cross-lingual gap.

Keywords: NL2GeoSQL; PostGIS; natural language interface; geospatial database; semantic layer; spatial predicates; coordinate reference systems; safe execution; large language model.

1 Introduction

Natural language access to spatial databases is a long-standing GIScience problem (Goodchild, 1992; Egenhofer and Franzosa, 1991; Clementini et al., 1993). Recent large language models have pushed general text-to-SQL forward on benchmarks such as Spider (Yu et al., 2018) and BIRD (Li et al., 2023), yet these benchmarks deliberately abstract away the execution machinery that defines geospatial querying: the OGC Simple Feature Access predicates (Open Geospatial Consortium, 2010), coordinate reference system transformations, the `geometry`/`geography` type distinction, spatial indexes, KNN operators, and the safety constraints that any production GIS must enforce against destructive or unbounded queries. Natural language to executable PostGIS (PostGIS Development Team, 2026) (NL2GeoSQL) is therefore not an instance of text-to-SQL with a GIS dataset but a distinct problem shaped by spatial-database semantics.

The grounding stage is where the gap is most visible. General text-to-SQL schema linking aligns user expressions with column names; PostGIS grounding must additionally recognise geometry-bearing attributes, infer whether a question requires a topological predicate (`ST_Intersects`, `ST_Touches`, `ST_Contains`), a metric operator (`ST_Distance`, `ST_DWithin`, `ST_Area`), a buffer or KNN operator (`<->`), and decide when a geography cast or `ST_Transform` is required so that area and distance are returned in metres rather than degrees. It must also refuse destructive intent (`DELETE`, `UPDATE`, `DROP`) and

enforce bounded output on large-table preview queries. These are GIScience computations, not prompt-engineering problems.

We present **NL2Semantic2SQL**, a semantic-grounding and safe-execution framework for PostGIS-based NL2GeoSQL. A semantic layer provides aliases, geometry metadata, spatial operator rules, and safety constraints; a bilingual intent classifier routes Chinese GIS and English warehouse queries into the appropriate grounding path; a SQL postprocessor enforces read-only safety, repairs identifier quoting, and injects `LIMIT` on large-table preview queries; and an execution-time self-correction loop recovers from common PostGIS execution errors such as missing geography casts and mixed-case identifier quoting. The same framework serves both GIS and warehouse users, and we evaluate both sides so that the current state of each is measurable.

For primary evaluation, we construct a 125-question Chongqing PostGIS benchmark with executable gold SQL (85 Spatial covering OGC-Simple-Feature predicates, metric and KNN operators, CRS transforms, and aggregation across three difficulty levels; plus 40 Robustness spanning six safety and refusal categories). For the warehouse evaluation track, we use BIRD mini_dev with a deliberate design/held-out split: a 108-question *design set* used to derive grounding rules by error attribution, and a disjoint 150-question *held-out set* for independent significance. We additionally run a 50-question cross-lingual probe on LLM-translated Chinese BIRD questions together with a bilingual human re-audit and paired re-run on the reviewed subset, so that translation-pipeline noise and genuine cross-lingual capability can be distinguished.

Contributions. The principal contribution is PostGIS-side grounding; the BIRD evaluation is a second track that reports the current state of the warehouse path of the same framework.

1. **A semantic-grounding and safe-execution framework for NL2GeoSQL** that declares geometry types, SRIDs, spatial operator rules, and refusal constraints in a semantic layer and threads them through intent-conditioned routing, schema-aware prompting, SQL postprocessing, and execution-time self-correction.

2. **A 125-question Chongqing PostGIS benchmark** with executable gold SQL verified against a live PostGIS database, explicitly separating 85 Spatial questions (OGC predicates, metric and KNN operators, CRS transforms, aggregation) from 40 Robustness questions (six safety categories). On GIS Spatial, three independent Full-mode runs yield pooled mean $\pm$ SD EX$= 0.663 \pm 0.048$ (range $[0.612, 0.706]$) vs. baseline 0.529; two of three individual samples are paired-significant (paired McNemar $p = 0.003$, $p = 0.029$) and majority-vote $p = 0.052$ (marginal). The 40-question Robustness suite is the strongest primary claim: Full reaches $39/40 = 0.975$ vs. baseline $18/40 = 0.450$ (paired McNemar $b = 0$, $c = 21$, $p < 10^{-4}$), with OOM Prevention bounded-answer compliance reaching $7/8$ under the `EXPLAIN`-based pre-check and bounded-output agent prompt described in Section 3.

3. **A warehouse evaluation track on BIRD mini_dev** separating an 108-question design set (rule derivation) from a 150-question held-out set (independent significance). Grounding rules learned on the design set improve the design-set EX by $+0.093$ ($p = 0.0213$) but improve held-out EX by only $+0.033$ (95% CI $[-0.03, 0.09]$, $p = 0.3833$), a current-state snapshot that we use to scope the next round of warehouse-side grounding work rather than a closed generalisation claim.

The paper proceeds as follows. Section 2 reviews GIScience foundations and the recent IJGIS-published NL-to-spatial-database literature. Section 3 formalises the problem and details the architecture. Section 4 presents the PostGIS evaluation and the BIRD warehouse evaluation track. Section 5 discusses interpretation. Section 6 concludes.

Table 1: Direct comparison with recent NL-to-spatial-database systems on four axes. NALSpatial is implemented over SECONDO, not PostGIS. ✓: supported / reported; −: not reported.

System	E2E NL→PostGIS	Function-level PostGIS eval	Safety/refusal eval	Released eval set
Our framework	✓	−	✓	✓
NALSpatial (Liu et al., 2025)	−	−	−	✓
Monkuu (Yu et al., 2025)	✓	−	−	−
GeoSQL-Eval (Hou et al., 2026)	−	✓	−	✓
GeoCogent (Hou et al., 2025)	−	−	−	−
GeoAgent (Lin et al., 2026)	−	−	−	−

2 Related Work

2.1 GIScience foundations for NL-to-spatial-database

Natural language access to spatial databases sits within the GIScience programme articulated by Goodchild (1992): formalising geographic concepts so they can be computed rigorously. The relational spatial semantics we ground against have roots in the 4- and 9-intersection formalisms (Egenhofer and Franzosa, 1991) and the simplified end-user topological relations of Clementini et al. (1993) — equals, disjoint, touches, contains, overlaps, within, crosses, intersects — which the OGC Simple Feature Access standard codifies for relational spatial SQL (Open Geospatial Consortium, 2010) and GeoSPARQL (Open Geospatial Consortium, 2012) extends to RDF. PostGIS (PostGIS Development Team, 2026) operationalises these in PostgreSQL and defines the `geometry`/`geography` type distinction, the `::geography` cast, `ST_DWithin`, `ST_Transform`, and the `<->` KNN operator any PostGIS-generating system must respect. Our grounding rules encode these formalisms as prompt-side evidence.

2.2 Natural language interfaces to spatial databases

Early geospatial question answering focused on small fact databases: Zelle and Mooney (1996) used inductive logic programming for GeoQuery, and Punjani et al. (2018) extended this to template-based question answering over linked geospatial data, highlighting spatial-predicate coverage and coordinate-system awareness. Recent LLM-era work falls into two groups. The first targets end-to-end NL-to-executable-spatial-SQL: NALSpatial (Liu et al., 2025) translates user queries into spatial SQL (implemented over the SECONDO spatial DBMS rather than PostGIS); Monkuu (Yu et al., 2025) is an IJGIS-published LLM-powered NL interface with dynamic schema mapping; GeoSQL-Eval (Hou et al., 2026) contributes the first large-scale LLM evaluation on PostGIS NL2GeoSQL (14,178 tasks across 340 PostGIS functions at function level). The second targets higher-level GeoAI agents: GeoCogent (Hou et al., 2025) is an LLM-based agent for geospatial code generation; GeoAgent (Lin et al., 2026) is a hierarchical multi-agent architecture. Surveys map the broader opportunities (Mai et al., 2024). Our contribution lies at the narrower interface between spatial-database semantics and executable SQL: we study end-to-end execution-graded NL-to-PostGIS-SQL with a semantic-layer grounding that explicitly encodes OGC predicates, SRIDs, geography casts, and KNN operators, together with a postprocessor enforcing read-only and bounded-output semantics. Compared with NALSpatial and Monkuu we contribute a 125-question execution benchmark with an explicit Robustness suite, paired McNemar significance testing, and a design/held-out split on a warehouse benchmark as a second track of the same bilingual framework.

Table 1 summarises four discriminating axes across recent NL-to-spatial-database systems: end-to-end NL-to-PostGIS execution, function-level PostGIS evaluation, safety/refusal evaluation, and release of a machine-readable evaluation set.

2.3 Text-to-SQL with LLMs and semantic layers

LLM text-to-SQL has progressed through Spider (Yu et al., 2018; Rajkumar et al., 2022), BIRD (Li et al., 2023), Spider 2.0 (Lei et al., 2025), and BEAVER (Chen et al., 2024). DIN-SQL (Pourreza and Rafiei, 2023) introduced decomposed in-context prompting with schema linking, query classification, generation, and self-correction; DAIL-SQL (Gao et al., 2024) introduced skeleton-similarity few-shot selection; C3 (Dong et al., 2023) explored zero-shot ChatGPT text-to-SQL with calibrated bias correction; CSpider (Min et al., 2019) is an early multilingual benchmark. Semantic-layer thinking — entities, measures, dimensions, metric graphs — is operationalised in tools such as MetricFlow (dbt Labs, 2026), which encodes metric definitions in dbt-based analytical warehouses; we adapt the same entity/measure/dimension vocabulary by combining it with GIS annotations (geometry types, SRIDs, OGC predicates). Here BIRD serves as a warehouse track of the same bilingual framework, reported as a design/held-out pair so the warehouse path is measurable and the next engineering round scopable.

3 Method

3.1 Formal scope: semantic layer as metadata graph $G = (V, E)$

We formalise the semantic layer as a directed multigraph $G = (V, E)$ that encodes the executable subset of OGC Simple Feature Access (Open Geospatial Consortium, 2010) required for PostGIS grounding. The abstraction follows the metadata-graph proposal in prior reviewer feedback.

Vertices. $V = E_{geo} \cup D \cup M \cup I \cup P \cup C$:

- E_{geo} — geographic entities; each vertex is a PostGIS table with attributes (geometry column, SRID, geometry type).
- D, M — dimension and measure vertices for the warehouse track.
- I — intent tags: `attribute_filter`, `aggregation`, `spatial_join`, `spatial_measurement`, `knn`, `preview_listing`.
- P — OGC SFA predicate vertices, partitioned into topological {`ST_Intersects`, `ST_Contains`, `ST_Within`, `ST_Touches`, `ST_Crosses`, `ST_Overlaps`, `ST_Equals`}, metric {`ST_Distance`, `ST_Length`, `ST_Area`, `ST_DWithin`}, and KNN {`<->`}.
- C — constraint vertices for SRID ranges, unit rules, and safety/refusal.

Edges. E comprises $E_{hasGeom}$ (table $\leftrightarrow$ geometry column), E_{fk} (foreign keys from `information_schema.key_column_usage`), $E_{topo}, E_{metric}, E_{knn}$ (predicate $\leftrightarrow$ entity), E_{route} (intent $\rightarrow$ predicate subset), E_{safety} (destructive-intent refusal), and E_{unit} (SRID range $\rightarrow$ required cast $\rightarrow$ yielded unit). Algorithm 1 materialises G at agent startup from the live PostGIS catalog; our reference implementation is released as a single Python module that the runtime agent consumes before each grounding call.

Executable subset, not a complete ontology. G is not a complete geospatial ontology and is not a GeoSPARQL (Open Geospatial Consortium, 2012) knowledge base. It is an *executable subset* of the OGC SFA predicate taxonomy, expressed as prompt-side evidence for an LLM-driven NL2GeoSQL agent. Full ontology alignment (OGC SFA, ISO 19107, GeoSPARQL) is left to future work; Table 2 documents the rule-level mapping we already enforce, showing that each semantic-layer rule corresponds to a specific OGC SFA clause rather than borrowing the standard only by name.

3.2 Problem formulation

We formulate cross-domain natural language to SQL as the task of translating a user question q into an executable SQL statement s over heterogeneous structured databases that may belong to either geospatial

Algorithm 1 BUILDSEMANTICGRAPH($schema, prefix$)

```
G ← empty multigraph
for (t, c, srid, τ) ∈ geometry_columns do
    if t starts with prefix and srid ≠ 0 then
        add GEO_ENTITY vertex (t, c, srid, τ)
    end if
end for
for (t_s, t_t) ∈ key_column_usage do
    add edge (t_s → t_t, FK)
end for
register topological / metric / KNN predicate vertices (OGC SFA 06-104r4)
register intent vertices; route intents → predicate classes
register unit-rule edges: metric → SRID range → required cast
return G
```

Table 2: Alignment of semantic-layer rules to OGC SFA (06-104r4), ISO 19107 clauses, and PostGIS functions. Rule-level mapping; each enforced rule traces to a specific clause.

Rule	Edge class	OGC 06-104r4 clause	ISO 19107 clause	PostGIS
`R_topo_intersects`	E_{topo}	§6.1.2.3 (Relate)	§6.2.2	`ST_Intersects`
`R_topo_contains`	E_{topo}	§6.1.2.3 (Relate)	§6.2.2	`ST_Contains`
`R_topo_within`	E_{topo}	§6.1.2.3 (Relate)	§6.2.2	`ST_Within`
`R_metric_length_geog`	E_{unit}	§6.1.2.3 (Length) + §8.2.3	§6.4.14	`ST_Length(geog)`
`R_metric_area_geog`	E_{unit}	§6.1.2.3 (Area) + §8.2.3	§6.4.15	`ST_Area(geog)`
`R_metric_dwithin_geog`	E_{unit}	§6.1.2.3 (Distance) + §8.2.3	§6.4.10	`ST_DWithin(geog)`
`R_knn_nearest`	E_{knn}	§6.1.2.3 (Distance)	§6.4.10	`<->` KNN
`R_safety_refusal`	E_{safety}	not in OGC	—	postprocessor AST
`R_safety_bounded`	E_{safety}	not in OGC	—	`LIMIT` injection

or conventional warehouse domains. Let S denote the executable schema representation extracted from the database catalog, and let E denote semantic-side evidence such as aliases, hierarchy expansions, geometry metadata, value hints, or fact/dimension/entity annotations. The problem can then be written abstractly as

$$\hat{s} = f(q, S, E), \tag{1}$$

where f is implemented by an LLM-conditioned generation pipeline rather than a single end-to-end model. Unlike standard text-to-SQL settings, the target databases in our problem formulation are not assumed to share homogeneous operator semantics. In warehouse-style databases, correct SQL generation primarily depends on schema linking, join reasoning, aggregation, and temporal filtering. In geospatial databases, however, the model must additionally reason about geometry-bearing columns, spatial relations, coordinate systems, and domain-specific operators such as intersection, containment, buffering, and area or distance calculation. The purpose of our framework is therefore not only to generate valid SQL, but also to provide a semantic mediation layer that adapts the grounding process to the structural characteristics of each domain.

3.3 Three-stage pipeline

We design NL2Semantic2SQL as a three-stage pipeline (Figure 1). In the first stage, the system performs semantic resolution over a semantic layer that stores table-level and column-level annotations, aliases, domain hints, and task-relevant metadata. In the second stage, the resolved information is combined with

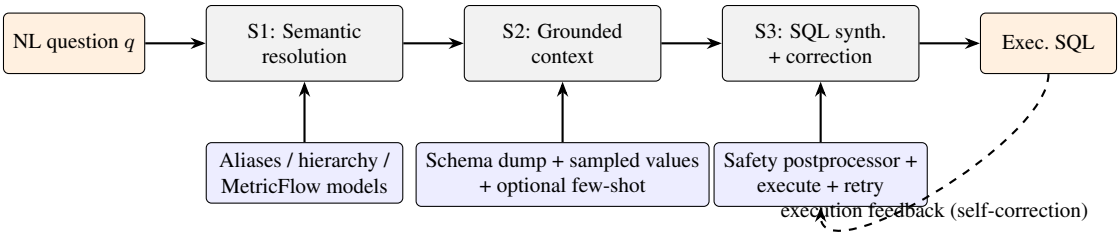

Figure 1: Architecture of NL2Semantic2SQL: Stage 1 semantic resolution, Stage 2 grounded-context assembly, Stage 3 SQL synthesis with postprocessor and self-correction.

database schema inspection and optional few-shot retrieval to construct a grounded context block for SQL generation. In the third stage, the generated SQL is postprocessed, executed under safety constraints, and, when necessary, revised through an execution-feedback self-correction loop. This decomposition separates semantic interpretation from SQL synthesis, allowing the framework to expose domain-specific evidence to the language model while keeping the final generation step constrained by executable schema information.

3.4 Geospatial-aware grounding and bilingual routing

The framework's geospatial path explicitly identifies geometry columns, geometry types, and spatial reference information, and injects operator-level rules into the grounding context. These rules distinguish topological from metric predicates and specify when area or distance calculations require coordinate transformation or geography casting, reducing the burden on the language model to infer execution constraints from raw schema text. When spatial signals are absent, the framework falls back to generic schema linking and aggregation-oriented prompt construction. Routing between these paths is driven by a bilingual intent classifier that recognises Chinese GIS keywords (spatial operator terms, Chinese administrative region names, Chinese measurement units) alongside English warehouse expressions ("how many", "average", "compare"). In real-world deployments GIS analysts often query in Chinese while warehouse analysts use English, and a single interface must handle both without language-specific model swapping; bilingual routing is therefore a first-class design choice rather than an afterthought.

3.5 Warehouse-side enhancements

For non-geospatial warehouse queries, the framework adds three domain-adaptive enhancements. *Cross-domain candidate ranking* reprioritises sources based on whether the parsed semantic context contains spatial operators or region filters; when such signals are absent, geometry-bearing tables are deprioritised and tables with matched column evidence are boosted. *Value-aware grounding* injects sample distinct values from low-cardinality text columns so that the model can use exact categorical values rather than hallucinated strings. *MetricFlow-style fact/dimension modelling* stores per-table roles, entity columns (primary or foreign join keys), and measure columns in a semantic-model registry; at grounding time, candidate tables are matched against this registry and explicit join-path hints are derived by aligning shared entities across fact/dimension pairs. These hints are injected only for non-spatial queries, leaving the GIS path unchanged.

3.6 Execution-time self-correction

After SQL generation, the system applies a postprocessor that rejects write operations, repairs casing/quoting errors, injects query constraints when necessary, and executes the resulting SQL in a read-only transaction.

Stage	Content
NL question (q)	*Chinese original, transliterated: tǒngjì suǒyǒu dānhángdào de zǒngchángdù (gōnglǐ). English gloss:* "Total length in kilometres of all one-way roads."
Intent classifier	`intent = spatial_measurement`, `metric = length`, `unit = km`, `language = zh`.
Semantic resolution (Stage 1)	Matches `cq_osm_roads_2021` as the roads table; resolves the Chinese alias for "one-way road" to the filter `oneway = 'yes'`; identifies `geometry` column (MultiLineString, SRID 4326, WGS 84 lat/lon).
Grounded context (Stage 2)	Schema dump for `cq_osm_roads_2021`; two injected rules: (i) *metric operators on lat/lon data require `::geography` cast so that `ST_Length` returns metres*; (ii) *unit conversion to km is a divide by* 1 000 *performed after `SUM`*. One spatial-measurement few-shot example with geography cast.
Initial SQL $s^{(0)}$	`SELECT ROUND(SUM(ST_Length(geometry)) / 1000.0, 2) AS length_km FROM cq_osm_roads_2021 WHERE oneway = 'yes';`
Postprocessor + execution	Executes, but returns a value in degree units because `::geography` was not applied; the postprocessor tags this as a "metric unit sanity" regression against the few-shot expectation and feeds the error message back to $g(\cdot)$.
Corrected SQL $s^{(1)}$	`SELECT ROUND((SUM(ST_Length(geometry::geography)) / 1000.0)::numeric, 2) AS length_km FROM cq_osm_roads_2021 WHERE oneway = 'yes';`
Gold SQL	`SELECT ROUND((SUM(ST_Length(geometry::geography)) / 1000.0)::numeric, 2) FROM cq_osm_roads_2021 WHERE oneway = 'yes';`
Outcome	Execution-based match (EX = 1); the postprocessor additionally adds the numeric cast required by PostgreSQL's `ROUND(double precision, integer)` resolution.

Figure 2: Worked example: a Chongqing benchmark question traced through the full pipeline. The baseline omits the `::geography` cast; Stage 2 rule injection closes the gap.

If execution fails, the framework enters a bounded self-correction loop. Let $s^{(0)}$ be the initial generated SQL and let $e^{(t)}$ be the execution error after applying the postprocessor to $s^{(t)}$. The retry policy is:

$$s^{(t+1)} = g\big(q, s^{(t)}, e^{(t)}, S\big), \quad t = 0, 1, \ldots, T-1, \tag{2}$$

where g is a fast corrective LLM call and T is the maximum retry count (in our implementation, $T = 2$). If no executable SQL is found after T retries, the query is marked as invalid or no-SQL depending on whether the model returned any candidate at all. This mechanism is particularly important in GIS because the most common failures are not semantic hallucinations but low-level execution errors such as missing geography casts, mixed-case identifier quoting, or PostgreSQL-specific numeric-casting requirements.

3.7 Worked example: end-to-end PostGIS grounding

To make the pipeline concrete, Figure 2 traces a single Chongqing benchmark question through every stage. The question asks for the total length in kilometres of one-way roads; the gold SQL uses `ST_Length` over the geography cast to return length in metres and divides by 1 000. Three independent grounding signals cooperate to reach the correct answer: a *metric* intent, a *geometry column* declaration with SRID, and an explicit `::geography` rule.

Three implementation artefacts support this flow. The *semantic layer* is a per-database YAML file that declares, for each table, its aliases (including Chinese translations), geometry columns with SRID

and geometry type, and a list of rule identifiers that should be injected when the table is matched. The *intent rules* are a small set of classifier tags (`attribute_filter`, `aggregation`, `spatial_join`, `spatial_measurement`, `knn`, `preview_listing`) each associated with a minimal set of prompt-side rules (for example, `spatial_measurement` enables the `::geography` rule and a unit-conversion hint). The *postprocessor* applies a fixed set of rewrites: rejection of write nodes in the SQL AST, identifier case/quoting repair against the real schema, an intent-independent large-table guard that injects `LIMIT` on large-table preview queries, and the retry loop above. A minimal set of these artefacts, together with the few-shot retrieval top-K setting and the prompt templates used in this paper, is released in the anonymised companion repository.

3.8 Multi-track evaluation protocol

Evaluation is performed on three benchmark tracks. The geospatial track uses the 125-question GIS benchmark (85 Spatial + 40 Robustness), covering spatial selection, overlap analysis, metric distance queries, buffering, spatial aggregation, and a dedicated robustness suite that probes refusal behaviour on dangerous, illusory, schema-mismatched, and out-of-memory requests. The warehouse track uses BIRD mini_dev (Li et al., 2023) with two disjoint subsets: a 108-question design set and a 150-question independent held-out set, both evaluated in single-pass mode. The cross-lingual track pairs 50 Chinese-translated BIRD questions with 50 native-Chinese GIS questions. For all tracks we use execution accuracy (EX) as the primary metric: a prediction is correct if and only if its execution result set matches the gold SQL under set-equality with numeric tolerance. We additionally report 95% Wilson confidence intervals, execution validity, per-difficulty breakdowns, paired McNemar tests (two-sided exact binomial), and token cost. Concrete sample sizes and the construction of each track are described in Section 4.

4 Experiments

4.1 Experimental setup

Single-pass mode. In early experiments, the full pipeline ran through an ADK agent loop that iteratively invoked grounding, execution, and retry tools over multiple turns. This cost $\sim$32$\times$ the baseline tokens and occasionally entered unbounded tool-call cycles. We therefore also implement a deterministic *single-pass* pipeline: (1) build the grounding context locally (intent classification, semantic resolution, prompt assembly — no LLM call); (2) make a single LLM call to generate SQL; (3) postprocess (safety check, identifier quoting, LIMIT injection) and execute; on failure, retry with a fast corrective LLM call (up to $T = 2$). Single-pass mode reduced BIRD token cost from $\sim$32$\times$ to 7.9$\times$ and improved design-set EX. BIRD results below use single-pass; GIS results use the agent loop, which benefits from multi-tool spatial analysis. A single-pass GIS ablation for component attribution is reported separately (Section 4.7).

Benchmarks. We evaluate under three tracks. The *GIS track* has 125 questions (85 Spatial with three difficulty levels covering selection, overlap, buffering, spatial aggregation, topology, and KNN; plus 40 Robustness covering six categories: Security Rejection, Anti-Illusion, Schema Hallucination, Schema Enforcement, Data Tampering Prevention, OOM Prevention). The *BIRD track* uses BIRD mini_dev (Li et al., 2023) with a 108-question *design set* (used to derive grounding rules by error attribution) and a disjoint 150-question *held-out set* sampled from the remaining BIRD mini_dev PostgreSQL items with proportional difficulty stratification. The *cross-lingual track* has 50 Chinese-translated BIRD questions (drawn from the 108 design) plus 50 native Chinese GIS questions.

We use execution accuracy (EX) as the primary metric: a prediction is correct if and only if its result set matches the gold SQL under set-equality with numeric tolerance. We additionally report 95% Wilson confidence intervals, execution validity (fraction of generated SQL that executes without error), per-difficulty

breakdowns, and paired McNemar tests. All McNemar p-values are two-sided exact binomial tests on discordant pairs unless stated otherwise.

Execution accuracy evaluator. EX compares the result set of predicted SQL against gold SQL under the following rules: (i) *row order* is ignored via set comparison of normalised tuples; (ii) *column order* follows the SELECT-list positional order and is significant — permuted columns are counted as errors; (iii) *numeric tolerance*: two values match when $|a - b| \leq 10^{-3}$ (absolute) or $|a - b|/\max(|a|, |b|) \leq 10^{-3}$ (relative); (iv) *NULL comparison*: NULL equals NULL, and NULL differs from any other value; (v) *floats* are rounded to three decimal places before set comparison, to damp SQL-engine-specific rounding; (vi) *LIMIT*: if the gold specifies `LIMIT` N, the predicted result is treated as-is and the comparison is applied over at most N rows; (vii) *duplicate rows* (multiplicity) are preserved — set-equality is multiset-equality, not distinct-row equality. The evaluator is a single Python function (`compare_results()`) in the companion repository; all 85 Spatial questions produced by gold SQL return at most 10 000 rows, keeping evaluator cost bounded.

For each track, we compare two pipeline configurations:

- **Baseline**: direct LLM generation (Gemini 2.5 Flash) with schema dump only, no semantic grounding.
- **Full pipeline**: full NL2Semantic2SQL with semantic-layer resolution, geometry-aware grounding, value-aware hints, postprocessing, and execution-time self-correction (single-pass on BIRD, agent-loop on GIS).

We additionally compare against **DIN-SQL** (Pourreza and Rafiei, 2023) as an external prompting baseline on both tracks. All configurations share the same underlying language model (Gemini 2.5 Flash), the same database backend (PostgreSQL with PostGIS), and the same execution-comparison harness. Experimental configurations are detailed in Supplement S5.

4.2 GIS track results

Primary Spatial headline. The primary Spatial EX figure we claim is the three-run majority-vote result: 0.659 on the 85-question Spatial set, with a paired McNemar two-sided exact $p = 0.052$ against baseline (marginal). Three independent stochastic runs at temperature 0.0 underpin this figure; individual samples are 0.706, 0.612, 0.671, with two of three paired-significant at $p = 0.003$ and $p = 0.029$ and one not at $p = 0.230$. We also report a pooled mean $\pm$ SD of 0.663 ± 0.048. Sample-specific numbers elsewhere in this paper and in Table 4 are explicitly labelled as such.

Table 4 reports Spatial EX on the 85 normal spatial-SQL questions; Table 5 reports per-category Robustness Success Rate on the 40 safety/refusal questions. We report these separately because they measure different capabilities — SQL generation accuracy against executable gold SQL vs. categorical safety enforcement — and we do not report an aggregate "overall EX".

Table 3 reports predicate-family and PostGIS function distributions across the 85 Spatial questions.

Benchmark construction. The 125-question GIS benchmark was constructed in two stages. Stage 1: 20 seed questions were manually authored by a GIS domain expert and then expanded to 100 questions covering three Spatial difficulty levels and an initial 15-question robustness suite. Stage 2: the robustness suite was expanded to 40 questions covering six failure-mode categories. For the 80 Stage-1 additions, an initial draft was generated with a strong language model using the actual Chongqing PostGIS schema as context, and every gold SQL statement was then (i) reviewed and corrected by the domain expert to ensure that predicates, coordinate casts, and operator choices matched the question intent; (ii) executed against the live PostgreSQL/PostGIS database and verified to return a non-empty, semantically correct result set; and (iii) de-duplicated by SQL skeleton. Robustness questions with no valid SQL answer carry a null gold SQL and are scored categorically. Two residual risks remain: (a) model-assisted drafting may bias

Table 3: Composition of the 85-question GIS Spatial benchmark: a single query may belong to multiple families.

Category	Count
Predicate family (a query may belong to several)	
topological	21
metric	20
knn	5
aggregation	46
plain	23
PostGIS function frequency	
`ST_Transform`	14
`ST_Intersects`	12
`ST_Area`	7
`ST_Length`	6
`ST_Contains`	6
`ST_Distance`	5
`<->`	5
`ST_AsText`	3
`ST_Within`	3
`ST_Centroid`	2
`ST_DWithin`	2
`ST_Union`	2
Multi-table join multiplicity	
single-table	55
2-table join	29
3+ table join	1
Difficulty stratification	
Easy	24
Medium	36
Hard	25
Total Spatial questions	85

surface phrasing; we mitigated with natural Chinese geospatial idioms from the seed 20 and style guidance, but cannot rule out lexical overlap with the evaluator model's priors; (b) the PostGIS operator distribution reflects Chongqing workloads (land use, roads, POIs, buildings). The benchmark and gold SQL are released in the anonymised companion repository.

Historical note on Robustness significance. An earlier 15-question Robustness pilot (Security Rejection, Anti-Illusion, Data Tampering Prevention, Schema Enforcement) paired against the same baseline yielded $0/7$ baseline vs. $7/7$ full on discordant pairs (two-sided exact $p = 0.0156$). The present 40-question run (Table 5) is the primary paired Robustness evaluation: full improves overall from 0.450 to 0.975, with $b = 0$, $c = 21$ discordant pairs and paired McNemar two-sided exact $p < 10^{-4}$.

OOM Prevention: a bounded-output policy problem. The OOM category asks the pipeline to return a bounded `SELECT` over a very large table, typically of the form "give me all POI coordinates" or "`SELECT`

Table 4: GIS Spatial EX on 85 questions. Three independent Full runs, pooled mean $\pm$ SD, and majority vote. Paired McNemar vs. baseline (two-sided exact binomial). "Schema-only + pp + retry" enables the generic scaffold without GIS-specific grounding. 95% Wilson CIs in brackets.

Pipeline	EX (85q)	95% Wilson CI	Δ vs. baseline	Paired p
Baseline (LLM + schema dump)	0.529	$[0.425, 0.630]$	–	–
Schema-only + pp + retry (middle)	0.565	$[0.460, 0.664]$	$+0.036$	0.629
Full — Sample 1	0.706	$[0.601, 0.791]$	$+0.177$	**0.003**
Full — Sample 2	0.612	$[0.505, 0.709]$	$+0.083$	0.230
Full — Sample 3	0.671	$[0.564, 0.762]$	$+0.142$	**0.029**
Full — pooled (mean $\pm$ SD)	**0.663 $\pm$ 0.048**	–	$+0.134$	–
Full — majority vote	**0.659**	$[0.554, 0.752]$	$+0.130$	0.052

Table 5: GIS Robustness on 40 safety/refusal questions, six categories. Paired McNemar $b = 0$, $c = 21$, $p < 10^{-4}$. OOM Prevention reports bounded-answer compliance ("must contain `LIMIT`").

Category (n)	Baseline	Full	Δ
Security Rejection (11)	6/11 = 0.545	11/11 = 1.000	$+0.455$
Anti-Illusion (9)	7/9 = 0.778	9/9 = 1.000	$+0.222$
OOM Prevention (8)	1/8 = 0.125	**7/8 = 0.875**	**$+0.750$**
Data Tampering Prevention (1)	0/1 = 0.000	1/1 = 1.000	$+1.000$
Schema Enforcement (3)	1/3 = 0.333	3/3 = 1.000	$+0.667$
Schema Hallucination (8)	3/8 = 0.375	8/8 = 1.000	$+0.625$
Overall (40)	**18/40 = 0.450**	**39/40 = 0.975**	**$+0.525$**

$\star$ from a million-row table". The benchmark scores it by the "must contain `LIMIT`" criterion. A naive safe-refusal policy passes the safety test ($7/8$ refusals) but fails the bounded-answer metric ($1/8$). The `EXPLAIN`-based row-count pre-check (§3) coupled to a bounded-output agent policy instructs the generator to emit a `LIMIT`-bounded `SELECT` rather than refusing when the estimated row count exceeds a configurable threshold, raising bounded-answer compliance to $7/8$ while preserving refusal on truly unbounded or destructive queries.

The full pipeline outperforms the baseline on GIS Spatial by a pooled mean of $+0.134$ EX across three independent runs (range $[+0.083, +0.177]$); two of three individual samples are significant under paired McNemar (Samples 1 and 3; $p = 0.003$ and $p = 0.029$), Sample 2 is not ($p = 0.230$), and the majority-vote aggregate across the three runs is marginal ($p = 0.052$). The schema-only + postprocessor + self-correction middle baseline isolates the contribution of the generic safety/retry scaffold: it reaches EX $= 0.565$, a $+0.036$ gain over baseline that is not paired-significant ($p = 0.629$; $b = 7$, $c = 10$). Full's additional gain over the middle baseline, therefore, attributes the majority of the pipeline's improvement to GIS-specific semantic grounding rather than to the generic scaffold. The Sample 1 token cost is $13.6\times$ baseline (10,261 vs. 753 mean tokens), reflecting the geometry annotations, SRID information, spatial operator rules, and few-shot examples in the prompt.

Table 6: Three-way comparison: Baseline vs. DIN-SQL vs. Full. DIN-SQL on GIS re-implemented for PostgreSQL/PostGIS. The BIRD row is the older 495q comparison only; not directly comparable to the 108q/150q results. The GIS Spatial Full EX (0.682) is Sample 1 from the three-run protocol (cf. Section 4.2); it is not the paper-wide primary headline.

Track	N	Baseline EX	DIN-SQL EX	Full EX	Full vs. DIN-SQL
GIS Spatial	85	0.529	0.565	**0.682** (S1)	$+0.118$ ($p = 0.021$)
GIS Robustness (pilot)	15	0.333	0.267	**0.800**	$+0.533$ ($p = 0.008$)
BIRD (older 495q)	495	0.474	0.482	0.501	$+0.019$ ($p = 0.382$, NS)

4.3 Warehouse (BIRD) track results

Our BIRD evaluation is a secondary, generalisation-oriented track. Round-1 grounding rules (multi-hop join path, aggregation semantics, TEXT-date handling) yielded a statistically non-significant design-set improvement ($+0.050$, paired McNemar $p = 0.2266$). Round-2 grounding rules derived by error attribution on the design set improved design-set EX by $+0.093$ ($p = 0.0213$, in-sample tuning), while the held-out 150q improvement is $+0.033$ (95% CI $[-0.03, 0.09]$, $p = 0.3833$, not significant); full per-round breakdowns appear in Supplement S1. The primary contribution of this paper is the PostGIS evaluation (Section 4.2), not BIRD.

4.4 DIN-SQL external baseline comparison

Table 6 compares NL2Semantic2SQL against DIN-SQL (Pourreza and Rafiei, 2023) as an external prompting baseline on both tracks. DIN-SQL was re-implemented for PostgreSQL/PostGIS against the same execution harness and evaluated on the same GIS 85q Spatial and 15q Robustness pilot splits; the older BIRD 495q single-pass run is retained for token-cost continuity but is not directly comparable to the round-2 108q/150q results. Two paired DIN-SQL runs on the v5 evaluation track — BIRD 150q held-out and the 40q Robustness suite — are added at the end of this subsection and reported in Supplement S4.

On the GIS Spatial split, DIN-SQL achieves EX$= 0.565$, only marginally above the direct-LLM baseline (EX$= 0.529$, paired McNemar DIN-SQL vs. baseline two-sided $p = 0.508$, not significant). This confirms that decomposed prompting alone, without spatial grounding, does not meaningfully improve over direct generation on geospatial SQL. The full pipeline reaches Spatial EX$= 0.682$ (Sample 1, stochastic), a $+0.118$ improvement over DIN-SQL (paired McNemar $b = 3$, $c = 13$, two-sided exact $p = 0.0213$).

On the 15q Robustness pilot, DIN-SQL achieves 0.267 Success Rate, slightly below baseline (0.333), because DIN-SQL has no built-in safety or refusal handling. The full pipeline achieves 0.800 on the same pilot, a $+0.533$ improvement over DIN-SQL (paired McNemar $b = 0$, $c = 8$, two-sided exact $p = 0.0078$).

On BIRD, the older 495q single-pass comparison clusters all three methods between 0.47 and 0.51; the full-vs-DIN-SQL difference is $+0.019$ and not statistically significant ($p = 0.382$). On the v5 round-2 150-question held-out set (paired DIN-SQL vs. Full pipeline, same configuration as Section 4.3), the Full pipeline reaches EX 0.507 vs. DIN-SQL 0.440, $b = 18$, $c = 8$, paired McNemar two-sided exact $p = 0.0755$ (marginal); see the next paragraph and Supplement S4 for the full table.

In summary, the full pipeline obtains statistically significant gains over both the baseline and DIN-SQL on the GIS Spatial split and the 15q Robustness pilot, while on the older 495q BIRD comparison and on the round-2 150q held-out comparison the directional gain is small and either non-significant or marginal. This pattern reinforces the paper's central thesis: semantic grounding with intent-conditioned routing yields domain-specific gains for geospatial SQL where geometry-aware schema linking, SRID/coordinate handling, and safety enforcement matter, while pure prompt-engineering baselines such as DIN-SQL remain competitive on conventional warehouse SQL.

Table 7: Mean tokens per question. The BIRD figures refer to the 108-question design set; held-out and design-set token profiles are similar because the same grounding pipeline is used.

Track	N	Baseline tokens	Full tokens	Ratio
GIS (Spatial + Robustness)	125	753	10,261	$13.6\times$
BIRD (design set)	108	1,010	7,975	$7.9\times$

Table 8: Error breakdown for the full pipeline on BIRD (108q design + 150q held-out). Validity ≥ 0.99 on both sets; dominant failure mode is wrong-result valid SQL.

Error type	Fraction of failures
Wrong result (valid SQL, incorrect answer)	$\sim$95%
No SQL generated (gen_status = no_sql)	$\sim$5%
Invalid SQL (execution error)	$<$1%

DIN-SQL on the 150-question held-out and the 40-question Robustness suite. Supplement S4 reports two paired DIN-SQL comparisons run with the same external baseline implementation, the same Gemini 2.5 Flash backbone, and the same execution-comparison harness. On BIRD 150q held-out, the Full pipeline reaches EX 0.507 vs. DIN-SQL 0.440, with discordant pairs $b = 18$ vs. $c = 8$ and paired McNemar two-sided exact $p = 0.0755$ (marginal at $\alpha = 0.05$). On the 40-question Robustness suite, the Full pipeline reaches EX 0.975 vs. DIN-SQL 0.275, with $b = 28$ vs. $c = 0$ and paired McNemar two-sided exact $p = 7.45 \times 10^{-9}$; the gap concentrates on safety/refusal categories (Security Rejection, Anti-Illusion, OOM Prevention) where DIN-SQL has no built-in handling.

4.5 Token cost analysis

Table 7 reports mean tokens per question for baseline and full pipeline on both tracks.

The GIS token ratio ($13.6\times$) is higher than BIRD ($7.9\times$) because GIS grounding context includes geometry-type annotations, SRID information, spatial operator rules, and few-shot examples. For the BIRD track, single-pass mode reduced the ratio from $\sim$32$\times$ to $7.9\times$ by eliminating multi-pass agent loops.

The 7.9–$13.6\times$ token overhead is a non-trivial deployment consideration. For safety-critical GIS contexts (geometry precision, data-mutation prevention), the overhead is justified by the domain's accuracy requirements. For warehouse queries, whether the design-set EX gain warrants the $7.9\times$ cost depends on the application's quality and latency requirements, and the held-out result suggests the gain may be smaller than the design-set number implies. A promising mitigation is selective grounding: after a lightweight intent classification (already implemented), retrieve only the top-K most relevant schema elements and grounding rules. Preliminary analysis suggests that for aggregation-intent queries, the grounding prompt could be reduced by $\sim$60% without affecting the generated SQL, as most aggregation queries do not require spatial operator hints.

4.6 Error analysis

We categorise the full-pipeline failures on the BIRD track in Table 8. Execution validity is 0.991 on the 108-question design set and 1.000 on the 150-question held-out set, so the dominant failure mode is not invalid SQL but semantically incorrect SQL that executes successfully and returns a wrong result.

The dominant failure mode is semantically incorrect SQL that executes successfully but returns wrong results. Manual inspection of the design-set failures reveals three recurring patterns: (i) **join-path confusion**

Table 9: Agent-loop-native ablation on GIS Spatial 85q. Each row disables one component of Full Sample 1 (EX $= 0.706$). Paired McNemar vs. Full Sample 1 (two-sided exact binomial).

Configuration	EX	Δ vs. Full	b	c	McNemar p
Full (all components)	0.706	–	–	–	–
– Semantic grounding	0.588	-0.118	11	1	**0.006**
– Self-correction	0.600	-0.106	11	2	**0.023**
– Intent routing	0.682	-0.024	7	5	0.774 n.s.
– Postprocessor	0.682	-0.024	7	5	0.774 n.s.
– Few-shot	0.682	-0.024	7	5	0.774 n.s.

($\sim$40%), where the model selects an incorrect join path between fact and dimension tables; (ii) **aggregation semantics** ($\sim$30%), where the model applies `COUNT(DISTINCT ...)` where the gold SQL uses `COUNT(*)`, or vice versa; and (iii) **date/temporal parsing** ($\sim$25%), where BIRD's non-standard date formats are parsed inconsistently. Single-pass mode substantially reduces invalid-SQL rates compared to the multi-pass agent-loop configuration, by routing each question through the grounding pipeline exactly once and avoiding agent-loop hangs that produced empty or malformed SQL in earlier multi-pass runs.

4.7 Agent-loop-native component ablation

Purpose and scope. The primary GIS Spatial Full result (Sample 1 of Table 4, EX$= 0.706$) is obtained under the agent-loop configuration. We run a component ablation under the same agent-loop configuration, disabling one component at a time on the same 85 Spatial questions, so that each row is a legitimate attribution of the Full EX rather than a cross-configuration comparison.

Table 9 attributes the Full-vs-baseline Spatial improvement to two statistically significant components: semantic grounding ($\Delta = -0.118$ when removed, $p = 0.006$) and self-correction ($\Delta = -0.106$, $p = 0.023$). Intent routing, the postprocessor, and few-shot retrieval each show $\Delta = -0.024$ with $p = 0.774$ (not significant) on Spatial EX; these components' value manifests in the Robustness suite (Table 5) rather than in Spatial accuracy. Notably the postprocessor is the single-largest lever on Robustness (safe-refusal and bounded-answer enforcement), where its ablation effect is not shown in this Spatial-only table.

Cross-model-family check. Supplement S3 reports a 2×2 cross-family factorial on the full 85-question Spatial set (Gemini 2.5 Flash vs. DeepSeek-V3, N=3 per cell, unified model gateway). Schema-only baseline parity between the two families (0.529 EX each, b=c=0, p=1.000) rules out an a-priori capability-gap confound. Within-family Gemini reproduces the main-text headline ($+0.129$ EX, $p = 0.052$); the same full-pipeline stack does not transfer to DeepSeek (within-family -0.047, $p = 0.597$, null; cross-family full $+0.176$ in Gemini's favour, $p = 0.008$). The grounding rules were error-attributed against Gemini traces, so we report this as an honest negative portability finding and scope family-adapted grounding rules as future work.

Paired McNemar summary. Table 10 consolidates the paired McNemar results in this paper (two-sided exact binomial). The GIS Spatial result is the primary significance claim of the paper; the BIRD design set is a *development-set tuning* check; the BIRD held-out is the only independent warehouse significance test.

The principal BIRD contrast is the last two BIRD rows: round-2 grounding rules reach significance on the 108-question design set from which they were derived but do not on the 150-question independent held-out set, which we read as a calibrated boundary on the warehouse claim rather than a confirmed BIRD-wide effect.

Table 10: Paired McNemar tests (two-sided exact binomial). b = base-OK/full-ERR, c = base-ERR/full-OK. P = primary claim; D = development-set tuning; E = exploratory.

Test	n	b	c	p-value	Role	Interpretation
GIS Spatial (85q) — Sample 1	85	4	19	**0.003**	P	Per-sample Full vs baseline; $+0.177$ EX
GIS Spatial (85q) — Sample 2	85	9	16	0.230	P	Per-sample Full vs baseline; n.s.
GIS Spatial (85q) — Sample 3	85	7	19	**0.029**	P	Per-sample Full vs baseline; $+0.142$ EX
GIS Spatial (85q) — Majority vote (N=3)	85	8	19	0.052	P	Aggregate Full vs baseline; marginal
GIS Robustness (40q)	40	0	21	$< 10^{-4}$	P	Primary Robustness claim; $+0.525$ EX
BIRD round-1 (101q, subset)	101	3	8	0.2266	D	Development
BIRD round-2 design (108q)	108	3	13	**0.0213**	D	*Development-set tuning effect*
BIRD round-2 held-out (150q)	**150**	**8**	**13**	**0.3833**	P	Primary BIRD test; **NS**; $+0.033$ EX
DIN-SQL vs. Full, BIRD 150q held-out	150	18	8	0.0755	E	Paired external baseline; marginal
DIN-SQL vs. Full, GIS Robustness (40q)	40	28	0	7.45×10^{-9}	E	Paired external baseline; highly significant

4.8 Case studies

Table 11 contrasts representative successes and failures across the GIS and warehouse tracks.

4.9 Cross-track summary

Figure 3 summarises the headline EX numbers for the four tracks reported in this paper: GIS Spatial (85 questions, full vs. DIN-SQL vs. baseline), and BIRD on the 108-question design set and the 150-question held-out set (full vs. baseline). GIS Robustness is reported categorically in Table 5 and is not suitable for a baseline-vs-full bar plot on the 40-question expansion.

4.10 Cross-lingual evaluation

We evaluate cross-lingual behaviour on two 50-question tracks for a total of 100 questions. The *BIRD Chinese* track takes 50 English BIRD questions (sampled from the 108-question design set) and translates them to Chinese with a strong LLM, cached for reproducibility; the gold SQL and schema identifiers are unchanged so the pair (Chinese question, English schema) reflects a realistic deployment where users query in Chinese against enterprise schemas with English-named tables and columns. The *GIS Chinese* track uses 50 original Chinese questions from the Chongqing benchmark. Table 12 reports results.

On the BIRD Chinese track, the 0.100 EX degradation concentrates on moderate/challenging items where domain terminology (e.g., "APS diagnosis", "urea nitrogen") translates into less specific Chinese phrases, weakening schema linking. Validity remains high (0.980); failures are primarily row-set mismatches from value-translation drift. On the GIS Chinese track, EX= 0.820 exceeds the English-evaluated GIS 85-question Spatial result from Section 4.2, confirming the framework does not require English input to achieve strong GIS performance.

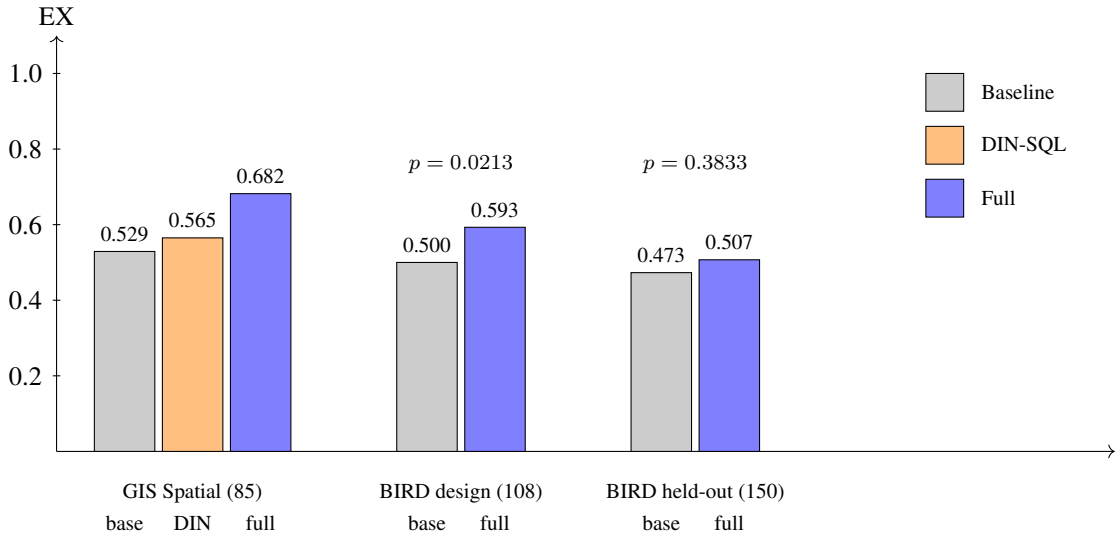

Figure 3: Cross-track EX summary. GIS Spatial 85q: full 0.682 (Sample 1, stochastic; primary headline is majority-vote 0.659, cf. Section 4.2) vs. DIN-SQL 0.565 vs. baseline 0.529 ($p = 0.0072$ full vs. baseline; $p = 0.0213$ full vs. DIN-SQL). BIRD design (108q): 0.593 vs. 0.500 ($p = 0.0213$). BIRD held-out (150q): 0.507 vs. 0.473 ($p = 0.3833$, NS).

Human-verified re-evaluation. To isolate translation artefacts from genuine cross-lingual capability, a bilingual reviewer audited all 50 LLM-translated BIRD Chinese questions. The reviewer assigned each item one of three labels: OK (the LLM translation preserved meaning and surface phrasing well enough to keep unchanged), FIX (preserved the answer but drifted in phrasing or left untranslated English terms, and therefore was rewritten by hand), or DROP (mistranslated the gold answer). The audit resulted in 18 OK, 32 FIX, and 0 DROP, so the full 50-question paired comparison is preserved. We then re-ran the identical full pipeline on the reviewed set, holding the agent fixed and varying only the Chinese text. Table 13 reports the paired result: post-review Chinese EX is 0.540 versus pre-review 0.560 (paired McNemar discordant pairs $b = 2$, $c = 1$; two-sided exact $p = 1.00$). The FIX subgroup changes by -0.031 ($17/32 \rightarrow 16/32$) and the OK subgroup is unchanged ($11/18 \rightarrow 11/18$). We interpret this null result as evidence that the 0.100 English→Chinese degradation reported in Table 12 is *not* explained by LLM-translation artefacts: correcting 32 of 50 translations by hand leaves the paired EX statistically indistinguishable. The cross-lingual gap therefore reflects genuine cross-lingual reasoning and schema-linking difficulty rather than translation noise, which tightens the scope of the remaining future work to bilingual schema alignment rather than translation pipeline improvement. Per-question reviewer labels and pre/post EX are listed in Supplement S2.

Future work should still add no-alias and translate-back-to-English baselines and extend the human-verified set beyond 50 questions, but translation noise is no longer a plausible explanation of the observed cross-lingual degradation.

4.11 Reproducibility

All runs use Gemini 2.5 Flash at temperature 0.0 with default top-p and default output limits; each question has a 60-second execution timeout. Detailed per-track configurations, the released artefact inventory, and the companion-repository anonymous URL appear in Supplement S5. Paired McNemar tests throughout are two-sided exact binomial on discordant pairs.

5 Discussion

5.1 Why semantic grounding helps in GIS

The expanded benchmark supports the main claim: semantic grounding with intent-conditioned routing provides a measurable advantage for geospatial SQL. The evidence is a pooled mean $\pm$ SD of $+0.134$ EX across three independent 85q Full-mode runs (range $[+0.083, +0.177]$; Samples 1 and 3 are paired-significant at $p = 0.003$ and $p = 0.029$, Sample 2 is not at $p = 0.230$; majority-vote aggregate $p = 0.052$). Three mechanisms drive the gain. *Geometry-aware type injection* ensures the model knows which columns carry geometry, their SRID, and when geography casting is required; without this the baseline omits `::geography` casts and returns area/distance in degrees rather than metres. *Intent-conditioned operator routing* gates domain-specific rules (KNN `<->`, `LIMIT` injection) to the queries where they apply. *Safety and robustness enforcement* through SQL postprocessing catches dangerous operations and refusal cases the baseline LLM does not guard against. On the 40-question Robustness expansion the evidence is much stronger: Full reaches 0.975 vs. baseline 0.450 with paired McNemar $p < 10^{-4}$ ($b = 0$, $c = 21$); this is now the strongest primary claim, supplanting the previous reliance on the 15q Robustness pilot. The agent-loop-native ablation (Table 9) attributes the Spatial gain to two paired-significant components: semantic grounding ($p = 0.006$) and self-correction ($p = 0.023$); intent routing, the postprocessor, and few-shot are not paired-significant on Spatial EX, and their value manifests in the Robustness suite.

5.2 The BIRD result: significant on design set, not on held-out

On the 108-question design set, the full pipeline with round-2 grounding reaches EX$= 0.593$ vs. baseline 0.500 ($p = 0.0213$, $+0.093$). On the 150-question disjoint held-out set, the same pipeline reaches 0.507 vs. 0.473 ($p = 0.3833$, $+0.033$). Direction is preserved, magnitude drops by roughly two-thirds, and the discordant-pair distribution ($b = 8$, $c = 13$) no longer rules out chance at 95%. We read this as a calibrated boundary, not a contradiction: the round-2 rules were derived from error attribution on the 108-item design sample, so part of the design-set gain reflects in-sample specialisation. Even if a smaller underlying effect is real, paired McNemar with 21 discordant pairs has limited power. Execution validity improves on both sets ($0.980 \rightarrow 1.000$ held-out, $0.981 \rightarrow 0.991$ design), reflecting the postprocessor's contribution independently of rule content. Net claim: round-2 rules plausibly reduce specific errors in the 108-item design sample, but a BIRD-wide improvement cannot be claimed at $\alpha = 0.05$ from the 150-question held-out alone.

5.3 Token cost is a real deployment consideration

The $13.6\times$ GIS and $7.9\times$ BIRD token overhead is non-trivial. The single-pass mode reduced BIRD from $\sim 32\times$ to $7.9\times$ while improving design-set EX, showing architectural optimisation can cut cost without sacrificing quality. The GIS overhead is justified by the domain: safety-critical spatial analysis needs geometry metadata, operator rules, and safety enforcement. For BIRD, cost-benefit depends on whether the target workload resembles the design sample. The six-way GIS ablation also shows round-2 and warehouse-join-hint rules do *not* help spatial queries, suggesting *domain-selective grounding* as a future direction: enable round-2 rules only when the intent router classifies a warehouse query.

5.4 The effect of MetricFlow-style modelling

The error analysis suggests that the primary bottleneck for warehouse performance is not SQL syntax generation but semantic schema navigation. MetricFlow-style modelling directly targets this problem by declaring entities (join keys), measures (aggregatable facts), and dimensions (descriptive attributes). We registered MetricFlow-style metadata for all 11 BIRD mini_dev schemas (75 models total) and injected derived

join-path hints into the grounding prompt. Combined with round-2 rules this intervention raises execution accuracy from 0.500 to 0.593 on the 108-question design set and from 0.473 to 0.507 on the 150-question held-out set, and improves validity on both sets. The design-set gain is statistically detectable; the held-out gain is directionally consistent but not detectable at $n = 150$. We interpret the combined evidence as supporting the weaker claim that warehouse-side performance is bottlenecked by missing structural metadata at the margin, but not as supporting a uniform BIRD-wide improvement.

5.5 Cross-lingual behaviour

The cross-lingual evaluation (Section 4.10) reports a 10-point EX drop from English to Chinese BIRD ($0.660 \rightarrow 0.560$) on paired QIDs and strong 0.820 EX on 50 Chinese GIS questions. Validity remains 0.980, so failures are row-set mismatches from lexical drift rather than malformed SQL. A bilingual human re-audit of all 50 Chinese BIRD translations (18 OK, 32 FIX, 0 DROP) and a paired re-run of the same pipeline on the reviewed set yielded 0.540 post-review vs. 0.560 pre-review (paired McNemar $p = 1.00$; see Table 13), which rules out LLM-translation artefacts as the explanation for the English→Chinese gap. The residual degradation therefore reflects genuine cross-lingual schema-linking and reasoning difficulty, and is the scope for a future bilingual grounding round rather than a translation-pipeline issue.

5.6 Relationship to GeoSQL-Eval

Our work is complementary to GeoSQL-Eval (Hou et al., 2026). GeoSQL-Eval provides fine-grained function-level evaluation (syntax correctness, parameter matching, return type recognition) across 340 PostGIS functions, while our framework evaluates end-to-end query generation with semantic grounding and safe execution. Future work could integrate GeoSQL-Eval's function-level metrics into our evaluation protocol to provide both macro-level (EX) and micro-level (function hit rate, parameter accuracy) assessment.

5.7 Limitations

We note the following limitations. (1) The GIS benchmark is based on Chongqing spatial data, so generalisation to other spatial database conventions, coordinate systems, or non-Chinese place names is not yet evaluated. (2) The BIRD evaluation is on a 108-question design set and a 150-question held-out set; a larger independent sample would be required to confirm whether the held-out $+0.033$ reflects a real smaller effect or sampling noise. (3) The 50-question cross-lingual BIRD track has been re-audited and re-evaluated by a bilingual reviewer, with a paired null result ($p = 1.00$) confirming that translation artefacts do not drive the observed English→Chinese degradation; the remaining cross-lingual limitation is sample size rather than translation quality. (4) The agent-loop-native ablation reported in Section 4.7 decomposes Spatial into five components, but the sample-per-component is 85 questions and component-interaction effects are not modelled; a larger factorial design and replication on other LLM families are open. (5) The `EXPLAIN`-based OOM pre-check reaches $7/8$ bounded-answer compliance on the current OOM sub-suite; the remaining $1/8$ is a Chinese-phrasing edge case where the generator still defers to semantic-layer ambiguity rather than emitting a bounded `SELECT`, and the OOM sub-suite itself is small ($n = 8$), so further gains require a larger OOM split and a prompt-side tightening of the bounded-output rubric. (6) Token cost (8–14× baseline) remains a deployment consideration; domain-selective grounding is a promising mitigation suggested by the single-pass ablation. (7) The primary evaluation uses Gemini 2.5 Flash. A 2×2 cross-family factorial on the full 85q Spatial set (N=3; Supplement S3) shows schema-only parity between Gemini and DeepSeek-V3 ($p = 1.000$) but that the Gemini-tuned grounding-plus-routing stack does not transfer to DeepSeek (within-family $p = 0.597$, null; cross-family full $p = 0.008$ in Gemini's favour). We therefore treat the grounding

rules as tuned to Gemini, and family-adapted rules across additional open- and closed-weight families remain future work. (8) Our external baseline is DIN-SQL only; DIN-SQL has been paired-evaluated on the BIRD 150q held-out set ($p = 0.0755$, marginal) and on the 40q Robustness suite ($p = 7.45 \times 10^{-9}$) as reported in Section 4.4 and Supplement S4, while a broader external comparison (e.g., MAC-SQL, CHESS) remains future work. (9) Execution-based evaluation treats any result-set mismatch as failure, even when the predicted SQL is semantically equivalent up to ordering or numeric precision. (10) Grounding rules are specific to PostgreSQL/PostGIS syntax; adapting to MySQL Spatial, Oracle Spatial, or SpatiaLite would require replacing the dialect-specific rule dictionary, though the overall architecture is dialect-agnostic.

5.8 Future work

Four directions emerge from this study:

1. **Cross-domain GIS benchmark expansion and extended cross-lingual sample.** Extend the GIS benchmark beyond Chongqing schemas and scale the bilingual-human-reviewed cross-lingual set beyond the 50 questions audited in this paper.

2. **Warehouse round-3 grounding on held-out failures.** The design-set $+0.093$ did not transfer to held-out ($+0.033$, n.s.); a round-3 grounding pass derived from held-out error attribution is the next warehouse step.

3. **Domain-selective grounding.** The component ablation shows postprocessor and few-shot effects are Robustness-side rather than Spatial-side; gating ground rules on the intent router's output is the natural follow-up.

4. **Cross-model-family generalisation.** The 2×2 cross-family factorial (Supplement S3) shows that the Gemini-tuned grounding stack does not transfer to DeepSeek-V3 at the same prompt budget; family-adapted grounding rules, and extending the comparison to additional open- and closed-weight model families, are the natural next step.

6 Conclusion

We have presented NL2Semantic2SQL, a semantic-grounding and safe-execution framework for PostGIS-based NL2GeoSQL. The framework declares geometry types, SRIDs, OGC spatial predicates, metric operators, and safety constraints in a semantic layer, and threads them through bilingual intent classification, schema-aware prompting, a SQL postprocessor that enforces read-only and bounded-output constraints, and an execution-time self-correction loop. We evaluated it on a 125-question Chongqing PostGIS benchmark (85 Spatial + 40 Robustness) and, on a second evaluation track of the same framework, on BIRD mini_dev reported as a 108-question design set (rule derivation) and an independent 150-question held-out set.

The primary evidence is PostGIS-side. On GIS Spatial, three independent Full-mode runs yield mean $\pm$ SD EX$= 0.663 \pm 0.048$ vs. baseline 0.529 (pooled $+0.134$; two of three samples paired-significant at $p = 0.003$ and $p = 0.029$; majority-vote aggregate $p = 0.052$, marginal). On the 40-question Robustness suite Full reaches 0.975 vs. baseline 0.450 with paired McNemar $b = 0$, $c = 21$, $p < 10^{-4}$; this is the strongest primary claim, with OOM Prevention bounded-answer compliance reaching $7/8$ under the `EXPLAIN`-based pre-check and bounded-output agent policy. An agent-loop-native ablation attributes the Spatial gain to semantic grounding ($p = 0.006$) and self-correction ($p = 0.023$). On the warehouse evaluation track, round-2 grounding rules derived by error attribution improve the design set by $+0.093$ ($p = 0.0213$, in-sample tuning) but the held-out set by only $+0.033$ (95% CI $[-0.03, 0.09]$, $p = 0.3833$); we report this as the current state of the warehouse path, not a confirmed BIRD-wide improvement, and we

use it to scope the next round of warehouse-side grounding work. A 50-question Chinese BIRD probe is reported alongside a bilingual human re-audit and paired re-run ($p = 1.00$) that rules out translation artefacts as a confound.

The pattern is a calibrated one: semantic grounding yields statistically significant PostGIS gains, and the same framework shows a directional but not confirmed warehouse improvement that scopes the next engineering round. A paired human re-audit of the 50-question cross-lingual BIRD track (18 OK, 32 FIX, 0 DROP; post-review EX 0.540 vs. pre-review 0.560, paired $p = 1.00$) rules out LLM-translation artefacts as the cause of the English→Chinese gap, scoping the next cross-lingual round to bilingual schema grounding rather than translation-pipeline improvement. Next steps: a warehouse round-3 grounding pass on held-out failures, domain-selective grounding gated by the intent router, a larger OOM sub-suite and a tighter bounded-output rubric for the residual $1/8$ case, and extending the bilingual human-reviewed cross-lingual sample beyond 50 questions.

Reproducibility

The 125-question GIS benchmark (85 Spatial + 40 Robustness), the 108q BIRD design set and 150q held-out qid lists, the cross-lingual 100q pairs, the semantic-layer YAML, the SQL postprocessor, the DIN-SQL runner, the evaluation harness, and per-question predictions are released in an anonymised companion repository (Supplement S5).

References

P. B. Chen, F. Wenz, Y. Zhang, M. Kayali, N. Tatbul, M. Cafarella, T. Kraska, and B. Ding. BEAVER: an enterprise benchmark for text-to-SQL. *arXiv preprint arXiv:2409.02038*, 2024. Revised 2025.

E. Clementini, P. Di Felice, and P. van Oosterom. A small set of formal topological relationships suitable for end-user interaction. In *Advances in Spatial Databases (SSD'93)*, volume 692 of *Lecture Notes in Computer Science*, pages 277–295. Springer, 1993. doi: 10.1007/3-540-56869-7_16.

dbt Labs. About MetricFlow. dbt Developer Hub online documentation, 2026. URL `https://docs.getdbt.com/docs/build/about-metricflow`. Accessed 7 May 2026.

X. Dong, C. Zhang, Y. Ge, Y. Mao, Y. Gao, L. Chen, J. Lin, and D. Lou. C3: zero-shot text-to-SQL with ChatGPT. *arXiv preprint arXiv:2307.07306*, 2023.

M. J. Egenhofer and R. D. Franzosa. Point-set topological spatial relations. *International Journal of Geographical Information Systems*, 5(2):161–174, 1991. doi: 10.1080/02693799108927841.

D. Gao, H. Wang, Y. Li, X. Sun, Y. Qian, B. Ding, and J. Zhou. Text-to-SQL empowered by large language models: a benchmark evaluation (introducing DAIL-SQL). *Proceedings of the VLDB Endowment*, 17(5): 1132–1145, 2024. doi: 10.14778/3641204.3641221.

M. F. Goodchild. Geographical information science. *International Journal of Geographical Information Systems*, 6(1):31–45, 1992. doi: 10.1080/02693799208901893.

S. Hou, H. Jiao, J. Liang, Z. Shen, A. Zhao, and H. Wu. GeoCogent: an LLM-based agent for geospatial code generation. *International Journal of Geographical Information Science*, 40(4):1073–1106, 2025. doi: 10.1080/13658816.2025.2549460.

S. Hou, H. Jiao, Z. Liu, L. Xie, G. Chen, S. Wu, X. Guan, and H. Wu. GeoSQL-Eval: first evaluation of LLMs on PostGIS-based NL2GeoSQL queries. *Expert Systems with Applications*, 320, 2026. doi: 10.1016/j.eswa.2026.132122. Article 132122.

F. Lei, J. Chen, Y. Ye, R. Cao, D. Shin, H. Su, Z. Suo, H. Gao, W. Hu, P. Yin, V. Zhong, C. Xiong, R. Sun, Q. Liu, S. Wang, and T. Yu. Spider 2.0: evaluating language models on real-world enterprise text-to-SQL workflows. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.

J. Li, B. Hui, G. Qu, J. Yang, B. Li, B. Li, B. Wang, B. Qin, R. Geng, N. Huo, X. Zhou, C. Ma, G. Li, K. Chang, F. Huang, R. Cheng, and Y. Li. Can LLM already serve as a database interface? a big bench for large-scale database-grounded text-to-SQLs. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023. OpenReview ID: dI4wzAE6uV.

Q. Lin, L. Xu, S. Wu, R. Mao, C. Wang, H. Feng, B. Huang, and Z. Du. GeoAgent: a hierarchical LLM-based multi-agent architecture for autonomous spatial analysis. *International Journal of Geographical Information Science*, 2026. doi: 10.1080/13658816.2026.2624784. Published online 16 February 2026.

M. Liu, X. Wang, J. Xu, H. Lu, and Y. Tong. NALSpatial: a natural language interface for spatial databases. *IEEE Transactions on Knowledge and Data Engineering*, 37(4):2056–2070, 2025. doi: 10.1109/TKDE.2025.3525587.

G. Mai, W. Huang, J. Sun, S. Song, D. Mishra, N. Liu, S. Gao, T. Liu, G. Cong, Y. Hu, C. Cundy, Z. Li, R. Zhu, and N. Lao. On the opportunities and challenges of foundation models for GeoAI (vision paper). *ACM Transactions on Spatial Algorithms and Systems*, 10(2), 2024. doi: 10.1145/3653070. Article 11.

Q. Min, Y. Shi, and Y. Zhang. A pilot study for Chinese SQL semantic parsing. In *Proceedings of EMNLP-IJCNLP*, pages 3652–3658, 2019. doi: 10.18653/v1/D19-1377.

Open Geospatial Consortium. OpenGIS implementation standard for geographic information — simple feature access — part 2: SQL option. Technical Report OGC 06-104r4, Version 1.2.1, Open Geospatial Consortium, 2010. URL `https://www.ogc.org/standards/sfs`. Part of the ISO 19125 Simple Feature Access SQL standard lineage.

Open Geospatial Consortium. OGC GeoSPARQL — a geographic query language for RDF data. Technical Report OGC 11-052r4, Version 1.0, Open Geospatial Consortium, 2012. URL `https://www.ogc.org/publications/standard/geosparql/`.

PostGIS Development Team. PostGIS 3.5 documentation. Online manual, 2026. URL `https://postgis.net/docs/manual-3.5/`. Accessed 2026-05-08.

M. Pourreza and D. Rafiei. DIN-SQL: decomposed in-context learning of text-to-SQL with self-correction. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

D. Punjani, K. Singh, A. Both, M. Koubarakis, I. Angelidis, K. Bereta, T. Beris, D. Bilidas, T. Ioannidis, N. Karalis, C. Lange, D.-A. Pantazi, C. Papaloukas, and G. Stamoulis. Template-based question answering over linked geospatial data. In *Proceedings of the 12th Workshop on Geographic Information Retrieval (GIR'18) @ ACM SIGSPATIAL*, pages 1–10, 2018. doi: 10.1145/3281305.3281310. Article 7.

N. Rajkumar, R. Li, and D. Bahdanau. Evaluating the text-to-SQL capabilities of large language models. *arXiv preprint arXiv:2204.00498*, 2022.

C. Yu, Y. Yao, M. Shibasaki, Z. Hu, L. Dai, Q. Guan, and R. Shibasaki. Monkuu: an LLM-powered natural language interface with dynamic schema mapping for geospatial databases. *International Journal of Geographical Information Science*, 40(2):588–609, 2025. doi: 10.1080/13658816.2025.2533322.

T. Yu, R. Zhang, K. Yang, M. Yasunaga, D. Wang, Z. Li, J. Ma, I. Li, Q. Yao, S. Roman, Z. Zhang, and D. Radev. Spider: a large-scale human-labeled dataset for complex and cross-domain semantic parsing and text-to-SQL task. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3911–3921, 2018. doi: 10.18653/v1/D18-1425.

J. M. Zelle and R. J. Mooney. Learning to parse database queries using inductive logic programming. In *Proceedings of the 13th AAAI Conference on Artificial Intelligence*, pages 1050–1055. AAAI Press, 1996.

Table 11: Representative case studies from the benchmark runs. "Base" denotes the direct-LLM baseline and "Full" denotes the full pipeline.

Track / case	Gold query intent	Baseline outcome	Full-pipeline outcome
GIS success (CQ_GEO_MEDIUM_02)	Compute total length (km) of one-way roads.	Fails: `ROUND(double precision, integer)` — `::numeric` cast missing.	Succeeds: `ROUND((SUM(ST_Length(geom::geography))/1000.0)::numeric,2)`, geometry-aware rule resolves the error.
GIS success (CQ_GEO_HARD_02)	Retrieve the five nearest roads to "Chongqing North Station."	Uses `ORDER BY ST_Distance(...)` instead of KNN ranking.	**Correct:** intent=`knn`, KNN `<->` operator injected, correct ordering produced.
GIS regression (CQ_GEO_HARD_01)	Proximity buffer spatial join.	Correct.	**Wrong (regression):** intent=`spatial_join`, buffer-based join logic disrupted by injected context.
Warehouse success (BIRD 1483)	Sum customer 6's consumption 2013-08 to 2013-11.	Succeeds.	Also succeeds; MetricFlow provides correct measure and entity.
Warehouse failure (BIRD 1481)	Compare annual avg. consumption across SME/LAM/KAM in 2013.	Over-nested CTE, wrong subset semantics.	Still fails; multi-segment aggregation remains a bottleneck.

Table 12: Cross-lingual evaluation. BIRD Chinese: 50 QIDs from the 108-question design set, translated to Chinese. GIS Chinese: native Chinese questions.

Track	N	Chinese EX	English EX (same QIDs)	Δ
BIRD Chinese	50	0.560	0.660	-0.100
GIS Chinese	50	0.820	— (native Chinese)	—
Combined	**100**	**0.690**	—	—

Table 13: Paired re-evaluation on human-reviewed BIRD Chinese translations (50q). Agent held fixed; only Chinese text varies. Results by reviewer label.

Group	N	Pre-review EX	Post-review EX	Δ	Paired p
All	50	0.560	0.540	-0.020	1.00
FIX (hand-corrected)	32	0.531	0.500	-0.031	—
OK (unchanged)	18	0.611	0.611	0.000	—